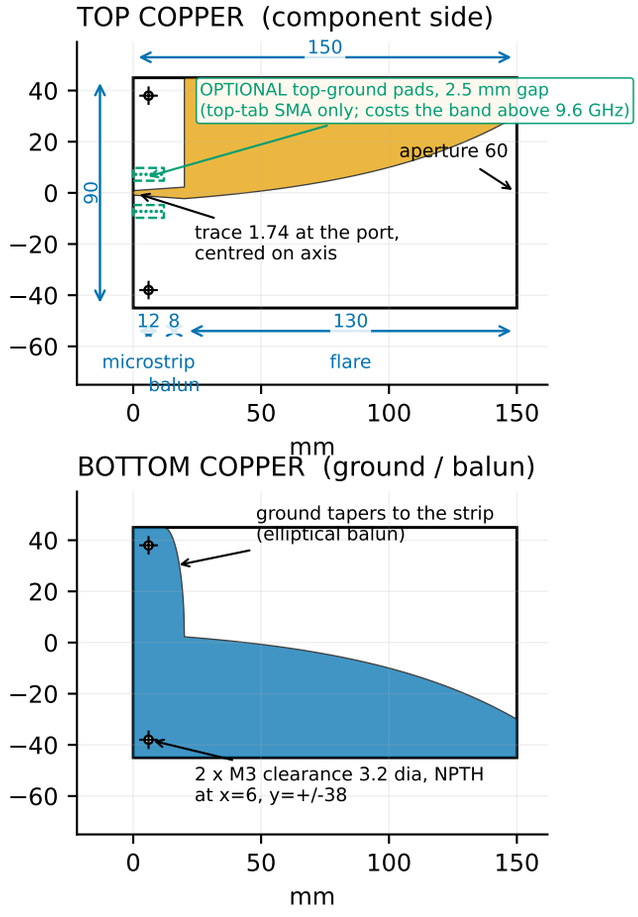

ANTIPODAL VIVALDI — 150 x 90 mm
Substrate RO4350B, eps_r 3.48, tan d 0.0037, thickness 0.762 mm. Copper 35 um (1 oz) both faces. Total stack 0.832 mm.
Taper y = A exp(0.0154 x') + B, growth a*T = 2.0, flare 106 mm over a 60 mm mouth (ratio 1.77).

SIMULATED PERFORMANCE (openEMS, model validated against the openEMS patch tutorial at Zin 49.7+4.5j):
|S11| < -10 dB over 2.48 to 12.00 GHz; 99.69 percent of 3-11 GHz below -10 dB, worst -9.98 dB at a single point (4.775 GHz).
Mean -16.8 dB across 3-11 GHz. Median Re(Zin) 45.6 ohm. Achieved by the tapered feed: 4.50 mm at the flare, 1.74 mm at the port.
Gain 7.02 dBi at 3 GHz, 11.9-12.5 dBi at 5-9 GHz, 9.3 dBi at 11 GHz. The 8-10 dBi target is met from about 4 GHz upward and NOT at 3 GHz; aperture-limited, not fixable by resizing.

CONNECTOR - qualified parts. Both were checked against their datasheets and both suit this 0.762 mm (0.030 in) board:
L-com RFPCB-SMA-FS18G-55 DC-18 GHz, specified for a 0.031 in board, 0.020 in pin, solder, 50 ohm, VSWR 1.2:1
Southwest 292-04A-9 DC-27 GHz, up to 0.065 in board, 0.010 in pin / 0.0635 in dielectric, SOLDERLESS clamp
The BOM asks for 18 GHz end-launch and warns off cheap 6 GHz parts. Either of these meets that.

NO EXTRA PADS ARE NEEDED. An end-launch lands its centre pin on the trace at the board edge and grounds its body to the plane on the opposite face. This board already presents both: the 1.74 mm trace runs to x=0 and the ground is full width there. The solderless part needs no soldering to the board at all.

The trace width is a good match rather than luck: 1.74 mm against the 0.0635 in (1.61 mm) dielectric diameter of the Southwest part - within 8 percent. A launch behaves best when the trace is close to the connector dielectric.

TOP-GROUND-TAB VARIANTS ARE SUPPORTED, with one condition. Such a connector grounds through tabs sitting on the top face either side of the pin, so it needs copper there. That turns the feed from microstrip into grounded coplanar waveguide and pulls the impedance down - unless the gap is wide enough that the coplanar coupling dies away. Simulated, not assumed (openEMS, same model as the antenna):

no pads (as drawn)	2.48 - 12.00 GHz	(9.52 GHz)	mean -16.8 dB	median Re(Zin) 45.6 ohm	[tapered feed]
pads, 1.0 mm gap	5.85 - 9.60 GHz	(3.75 GHz)	mean -14.2 dB	median Re(Zin) 39 ohm	[uniform feed*]
pads, 2.5 mm gap	4.35 - 9.60 GHz	(5.25 GHz)	mean -14.7 dB	median Re(Zin) 41 ohm	[uniform feed*]

* The pad rows were simulated on the earlier 1.74 mm uniform feed and have NOT been re-run with the taper. Their upper limit comes from the pads, not the feed, so the 9.60 GHz ceiling is expected to carry over; the low end will not.

PADS ALWAYS COST BANDWIDTH AT THE TOP OF THE BAND. Widening the gap to 2.5 mm recovers the low end - the coplanar coupling that pulls the impedance down dies away past about three substrate heights - but the upper limit stays at 9.60 GHz either way, because at 12 GHz a 2.5 mm gap is a fifth of a wavelength and the pads act as a discontinuity.

SO: prefer a bottom-ground or clamp part and omit the pads entirely. If a top-tab part must be used, use a gap of 2.5 mm or more and accept losing the band above 9.6 GHz. The pads are drawn dashed and carried on DXF layers COPPER_TOP_OPT_PADS and VIA_OPT so they can be omitted. They MUST be stitched to the ground plane by the vias shown; unstitched copper beside a feed line is a resonator, not a ground.

DXF layers: BOARD_OUTLINE, COPPER_TOP, COPPER_BOTTOM, DRILL, and the optional COPPER_TOP_OPT_PADS + VIA_OPT.
Units mm. Origin at the feed edge, y=0 on the antenna axis.